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Appendix

Section 1: Data and Methods

1.1: Data Sources

For our analytic sample of 271 drugs that experienced first generic entry between 2011 and 2023, we constructed a longitudinal dataset containing each drug's total spending, prescription volume, rebates, and retail and net prices for each calendar year from 2010 to 2024. We constructed that panel using data from the following sources:

1. Part D Prescription Drug Events (PDE) 100% file, 2009-2024. The PDE file contains a record for every prescription drug purchase covered by a Part D plan during the year, analogous to 100% claims files for other medical services. We used these data to estimate total prescription volume, average retail prices, and total spending at retail prices (not accounting for rebates) for each drug. We also used the data to calculate total coverage gap discounts, which we used along with manufacturer rebates to calculate net spending and net prices, as defined in the paper. Coverage gap discounts were point-of-sale discounts that manufacturers were obligated by law to extend to purchasers for some brand-name drugs dispensed to Part D enrollees in the third phase of the Part D benefit (known as the coverage gap, or "donut hole").
2. Part D rebate data furnished by CMS, 2010-2024. These data are generally unavailable to researchers. They contain total annual manufacturer rebates to Part D plan sponsors (the insurers that offer Part D plans), reported at the drug-by-plan level (where drugs are identified by 11-digit National Drug Codes (NDCs). As with the PDE, the rebate data are available in a consistent format for both standalone Part D plans (PDPs) and Medicare Advantage Part D plans (MA-PDs). In addition to total annual manufacturer rebates, the file also includes total other direct and indirect remuneration (DIR) received by the plan, principally from pharmacies (known as "pharmacy DIR"). We used the data on manufacturer rebates to calculate annual net spending and net prices by drug, but we did not use the data on other DIR payments.
3. Micromedex Expanded Red Book dataset, March 2026. The Red Book is a commercially available dataset maintained by Merative. The file contains a set of drug characteristics and price measures for each NDC. We used the drug characteristics in the data to group NDCs together as "drugs," for both our main specification and an alternative specification. We also used the Red Book data in conjunction with other sources to identify NDCs as either brand-name, authorized generic, or generic.
4. Medicaid Drug Rebate Program (MDRP) data, downloaded February 2025. This is an NDC-by-quarter dataset, downloadable from Medicaid.gov and containing information on drugs included in the Medicaid Drug Rebate Program. The data elements we used were the date that each NDC was first marketed, and whether it is marketed under a New Drug Application (NDA), an Abbreviated New Drug Application (ANDA), a Biologics License Application (BLA), or other.
5. Comprehensive NDC SPL Data Elements File (NSDE), downloaded March 2026. This is an NDC-level database downloadable from FDA. We used the data to supplement the MDRP data,

because the NSDE includes each NDC's marketing status and first marketing date, and their values are missing less commonly in NSDE than in MDRP files.

6. Paragraph IV Patent Certification List, version dated April 13, 2026. This document, published by FDA, contains a list of brand-name drugs whose patents have been challenged by generic drug manufacturers. Along with the name and NDA license number of each brand-name drug, the relevant variable is first commercial marketing date by a generic manufacturer eligible for a 180-day generic drug exclusivity from FDA.[1]

1.2: Sample Composition and Identifying Year of First Generic Entry

In this analysis, we define a drug as the set of 11-digit NDC codes that share the same active ingredient or combination of active ingredients. Under this broad definition, different dosage forms, strengths, and routes of administration are pooled together if they share the same active ingredients. To group NDCs into drugs using this definition, we used the Generic Cross Reference (GCR), an NDC-level variable included in the Red Book database.[2]

Our definition of "drug" pools different manufacturers, as well as across dosage forms and other characteristics. So, in a particular year, a single drug can potentially have brand-name sales for which we can calculate an average brand-name price; authorized generic sales for which we can calculate an authorized generic price; and generic sales for which we can calculate a generic price.

We constructed our analytical sample of small-molecule drugs experiencing first generic entry between 2011 and 2023 by following these steps:

1. To limit the sample to small-molecule drugs, we excluded all drugs for which any NDC was indicated in the MDRP or NSDE files as being licensed under a Biologics License Application (BLA).
2. We identified all drugs that had no generic sales in the 2009 or 2010 PDE data, but that did have generic sales in a subsequent year, and that also had brand-name sales before the year in which the first generic sales occur. This step identified 352 candidate drugs.
3. Wanting to ultimately include only drugs that had a total sales volume of at least 100 standardized prescription units (defined below) in the year before first generic entry, we dropped 24 drugs out of our initial 352 because they never reached 100 units in any single year in our data.
4. Among the drugs identified in the previous step, we compared the year that generic versions first appeared in the PDE data to the first generic marketing dates reported in the MDRP, NSDE, and Paragraph IV certification datasets. We applied each of the conditions below in order for each drug until one of the conditions was met. Applying this procedure led us to drop 15 drugs from the sample and assign a first generic entry year to each of the 313 remaining drugs. For clarity in describing these steps, let first_appearance denote the year that generic sales first appear in the PDE in 2011 or later (identified in the previous step), and let entry_year denote the year of first generic entry assigned to each drug following this procedure.

[1] For the latest data and more information, see www.fda.gov/drugs/abbreviated-new-drug-application-anda/patent-certifications-and-suitability-petitions.

[2] In rare instances, the Red Book assigns different GCR codes to NDCs that contain the same combination of active ingredients. In those cases, we consolidated the GCR codes so that all NDCs sharing the same set of active ingredients are grouped together.

a. If first_appearance equals the year of each (non-missing) generic entry date sourced from another dataset (MDRP, NSDE, and Paragraph IV certification list), then set entry_year = first_appearance.
b. If no generic entry dates are available from the MDRP, NSDE, or Paragraph IV certification files, then set entry_year = first_appearance.
c. If the generic entry date reported in either the MDRP or NSDE is earlier than 2011, then drop the drug from the sample. (This pattern indicates that generic versions of the drug were previously available, but off the market in 2009 and 2010 before re-appearing in a later year. For these drugs, first generic entry occurred outside the study period.)
d. If the MDRP or NSDE generic entry dates indicate that the first generic launched in November or December, and first_appearance is the following year, then entry_year = (first_appearance – 1).
e. If the PDE data indicate no generic sales in the year (first_appearance + 1), and the MDRP and NSDE indicate first generic entry at least two calendar years later than the first_appearance, then the entry_year is the second year that generic sales appear in the PDE (two or more years later than first_appearance).
f. If first_appearance equals the first generic entry year in the MDRP and NSDE data (but not the Paragraph IV certification data), then the entry_year = first_appearance.
g. If the first_appearance is later than the first generic entry year indicated in MDRP or NSDE, then entry_year = first_appearance.
h. If none of the above criteria are met, then drop the drug from the sample. This excludes drugs for which first_appearance cannot be reconciled with the generic entry dates reported in MDRP and NSDE.)

5. Drop from the sample all drugs with fewer than 100 standardized prescription units (defined below) in the year before first generic entry. This reduces the number of drugs in the sample from 313 to 301.
6. Among the remaining sample, identify and drop any drugs whose brand-name sales prior to first generic entry were split between *both* multiple product brand names *and* multiple brand-name manufacturers. We counted brand names using only the first word of the product name provided in the Red Book, and we excluded NDCs associated with over-the-counter versions, product repackagers, and authorized generics. The purpose of this step is to identify and exclude rare cases in which multiple brand-name drugs, made by different manufacturers, share the same combination of active ingredients. An example of such a drug is bromfenac sodium, which was sold under the brand names Prolensa (made by Bausch & Lomb) and Bromday (made by Sun Pharmaceuticals). Applying this data check reduced the number of drugs in our sample from 301 to 271 (our final sample size).

1.3: Calculating Standardized Units of Volume

Our measure of prescription volume is a 30-day supply of the drug, with an adjustment to account for drugs that are intended to be taken for fewer than 30 days. The number of standardized prescription units is calculated for each claim appearing in the PDE file, on the basis of the reported days supply corresponding to the prescription fill (represented in the variable days_supply). Each claim was assigned a number of standardized prescription units equal to days_supply divided by 30, but not less than one. Under that definition, a 90-day prescription fill for a statin corresponds to 3 standardized units, and a prescription for a 14-day course of antibiotics counts as one standardized unit.

1.4: Final Sample Composition

After applying the steps outlined above to identify drugs experiencing first generic entry between 2011 and 2023, and the year that first generic entry occurred for each drug, our final analytic sample included 271 drugs. Table S-1 reports the number of drugs per year, the number of 30-day-equivalent prescriptions they represented in their last year before first generic entry, and the share of total brand-name prescriptions in Part D that they represented.

Table S-1: Drugs Included in Analysis

Year of First Generic Entry	Drug Count	Prior Year Scripts (millions)	Share of Total Brand Scripts in Prior Year
2011	23	26.3	5.8%
2012	22	32.0	7.3%
2013	22	11.8	2.9%
2014	23	23.0	5.3%
2015	20	30.2	7.1%
2016	26	21.5	5.1%
2017	23	5.7	1.4%
2018	17	3.7	0.9%
2019	22	11.5	3.0%
2020	22	2.9	0.8%
2021	13	4.7	1.3%
2022	23	9.7	2.7%
2023	15	11.8	3.3%

1.5: Sample Weights

All of the estimates presented in the report's figures are weighted by a measure of total pre-generic-entry prescription volume. Specifically, the sample weight assigned to each drug is equal to its total 30-day-equivalent prescriptions in the year before first generic entry, expressed as a percentage of all prescriptions for brand-name drugs in that year. Specifying the weights this way allows us to assign greater weight to more widely used drugs without assigning greater weight to drugs experiencing generic entry later in the sample period simply because of increasing enrollment and utilization in Part D over time.

1.6: Seasonality of Generic Entry Date

In each drug's first generic entry year (denoted 0 in the figures), generic drugs appear in the data for less than a whole year. We checked for any systematic patterns in the timing of first generic entry across months of the year. We recorded each drug's first generic entry date using the MDRP data and NSDE data. The two data sources gave different answers in some cases, but neither produced results suggesting a systematic pattern in the timing of first generic launch dates within the year.

Table S-2: Timing of First Generic Entry Within the Year

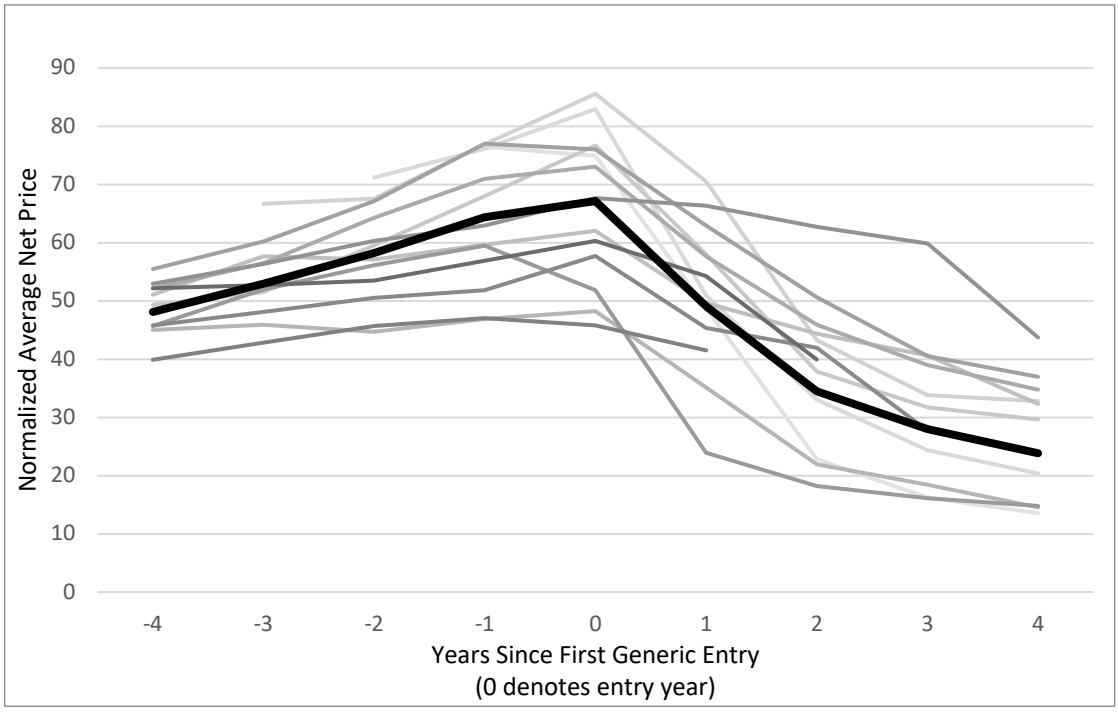
	Date Sourced from MDRP File	Date Sourced from NSDE File
Average Number of Months that First Generic Entrant is On the Market		
Weighting by Number of Drugs	6.2	6.1
Weighting by Drugs' Sample Weights	6.5	6.1
Distribution of Entry Dates by Calendar Quarter (%)		
Jan – Mar	23.6	24.0
Apr – Jun	22.9	24.4
Jul – Sep	30.2	29.5
Oct – Dec	23.2	22.1

Section 2: Additional Analyses

2.1: Heterogeneity by Generic Entry Year Cohort

The figure below depicts the aggregate change in total spending at net prices, for the entire sample (in black) and for each generic entry year cohort from 2011 through 2023 (in gray). The black line depicting average net price by year for the entire sample corresponds to Figure 3 in the report. Average net prices fell following first generic entry for every cohort of drugs.

Figure S-1: Normalized Net Price by Year of First Generic Entry



2.2: Heterogeneity by Tercile of Pre-Generic-Entry Spending

The article includes results stratified by tercile of total spending in the last year prior to first generic entry. Below are figures corresponding to Figure 1 and Figure 2 in the text, but specific to the drugs in each spending tercile.

Figure S-2: Normalized Prices by Tercile of Pre-Generic-Entry Spending

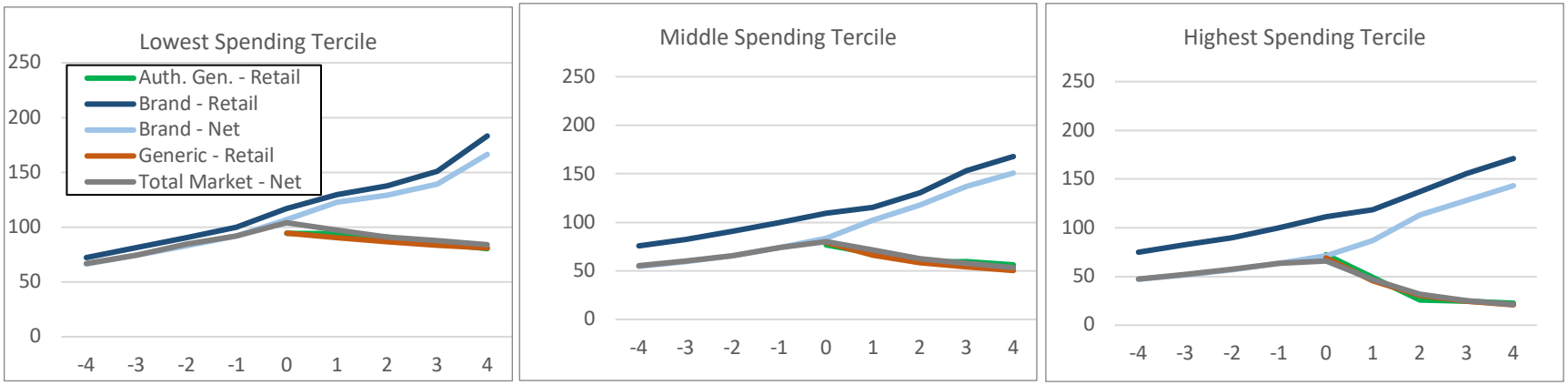
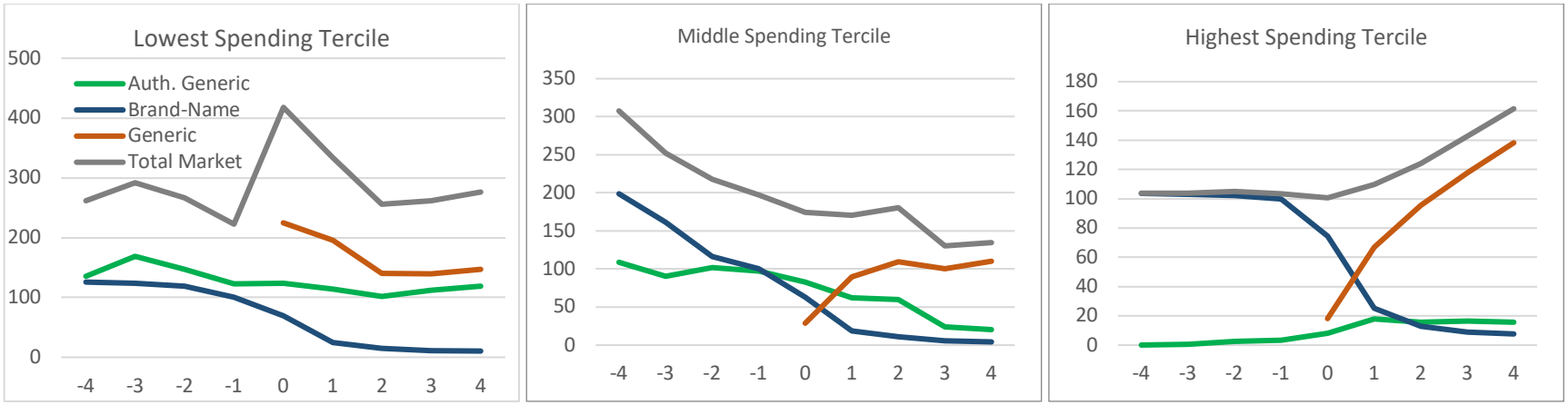


Figure S-3: Normalized Prescription Volume by Tercile of Pre-Generic-Entry Spending



2.3: Results Defining Drugs More Narrowly

For the figures in this section, NDCs are only grouped together as the same drug if they share the same formulation and route of administration (as identified in the Red Book) in addition to the same combination of active ingredients. This alternative definition of a drug is narrower than the definition used elsewhere. For example, under this alternative definition, capsule, tablet, and extended-release tablet formulations of the same active ingredient (all taken orally) are treated as different drugs, and a tablet version that is dissolved sublingually would be yet another different drug. This alternative definition is closer to the scope of an NDA or ANDA, in that different strengths of the same formulation of the same active ingredient are covered under the same NDA or ANDA, but different ingredients, different formulations, and different routes of administration usually require separate NDAs and ANDAs.

Using this definition, we identify 452 drugs that first experienced generic entry between 2011 and 2023. Whether using this narrower drug definition or the broader definition used for our main results, the results are very similar. For instance, using the narrower definition, net brand prices increased 94 percent and generic prices fell 66 percent, on average, over the four years following first generic entry. Using the broader drug definition, the corresponding changes in average prices were 99 percent for brand-name drugs and -67 percent for generic drugs.

Figure S-4: Average Aggregate Prices Using Narrower Drug Definition

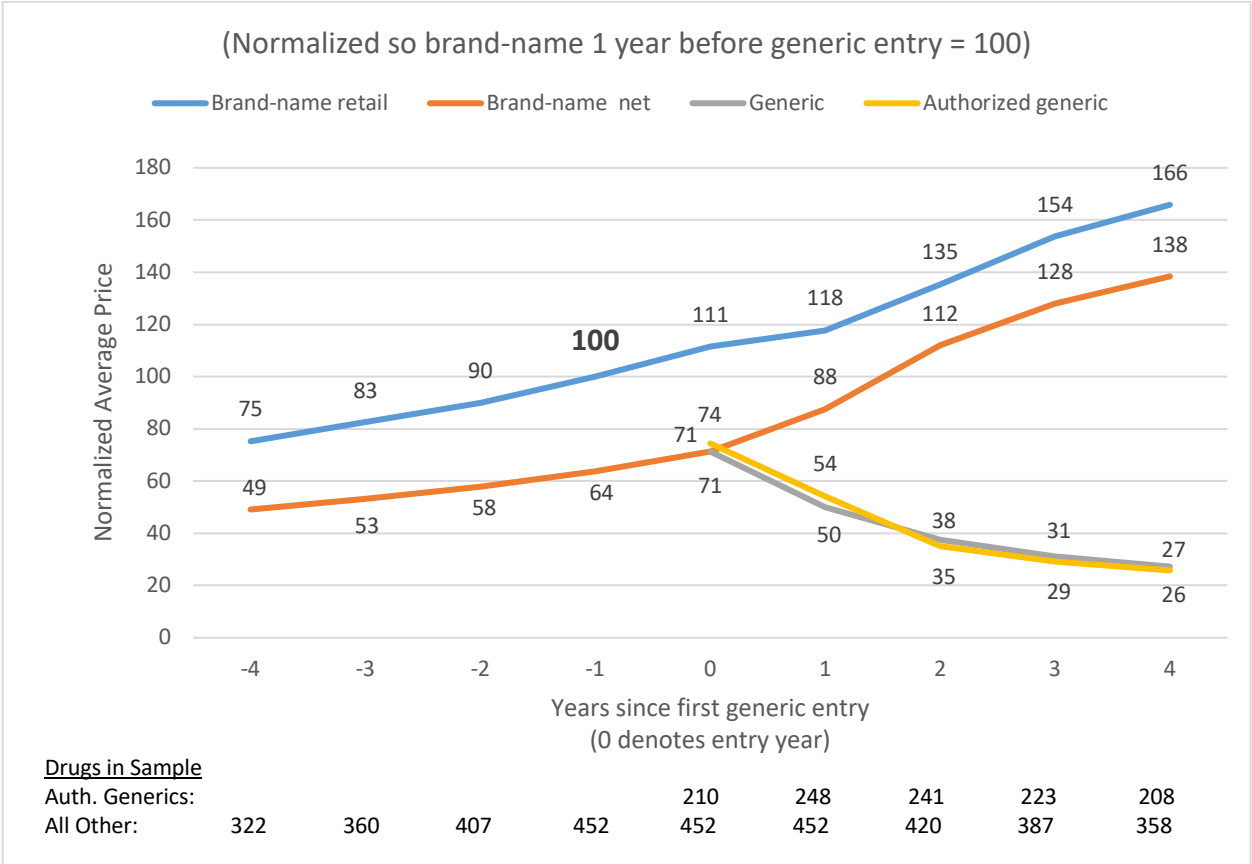


Figure S-5: Standardized Prescriptions Using Narrower Drug Definition

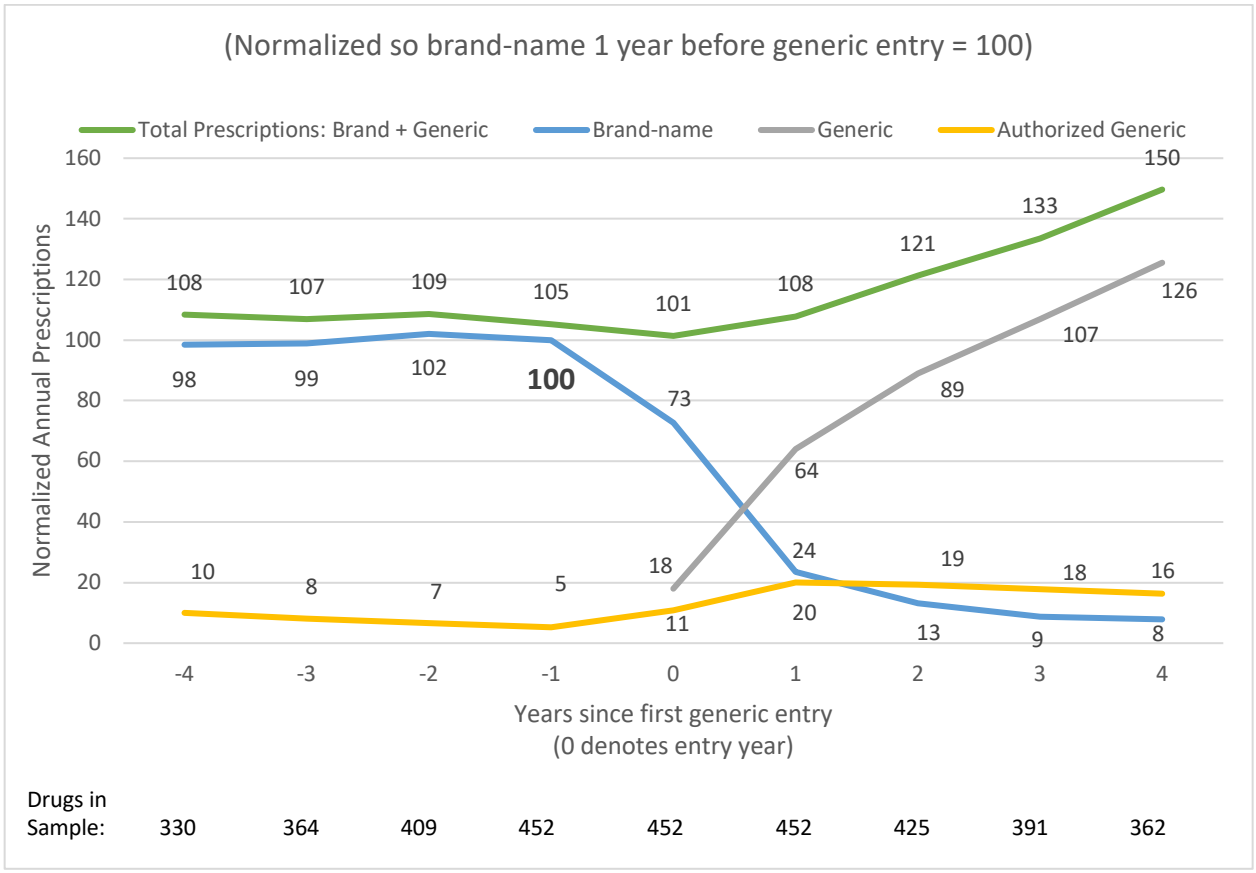
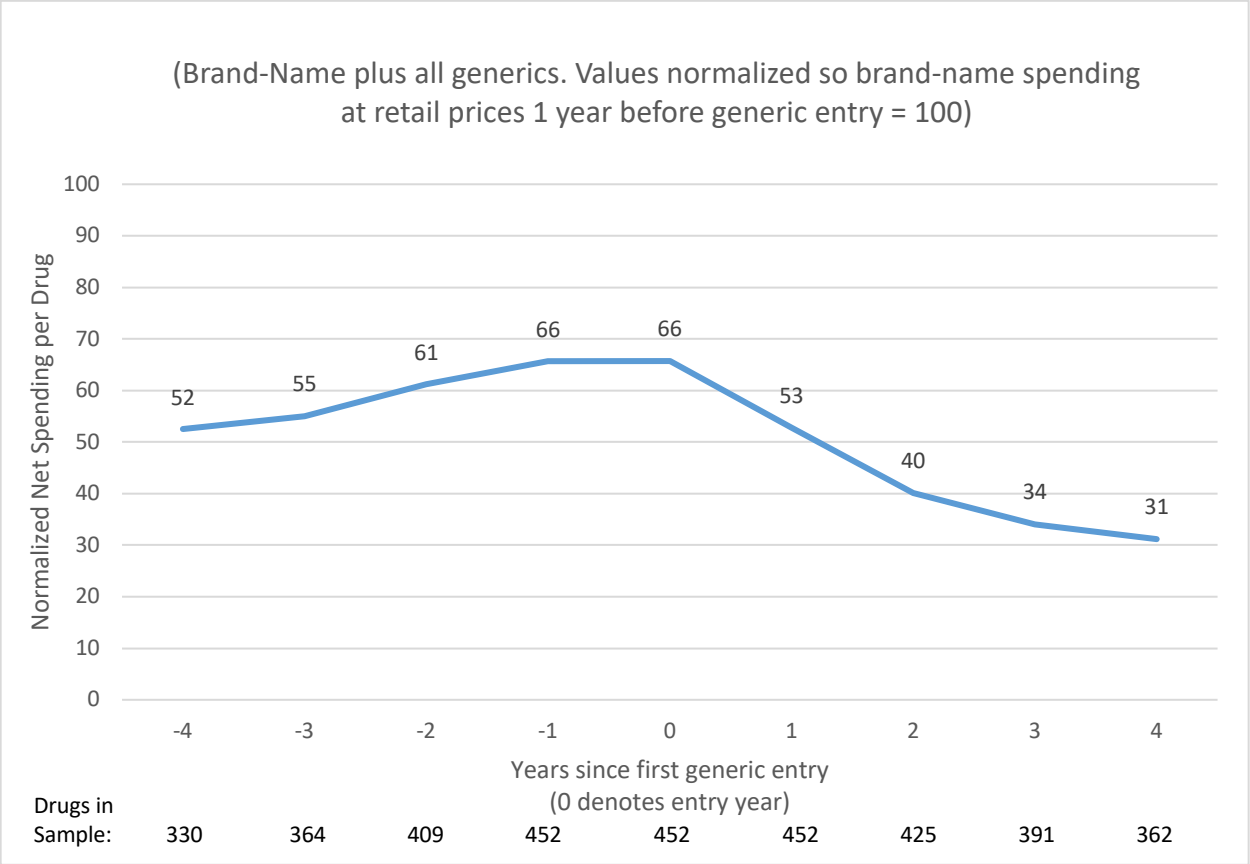


Figure S-6: Part D Net Spending Per Drug Using Narrower Drug Definition



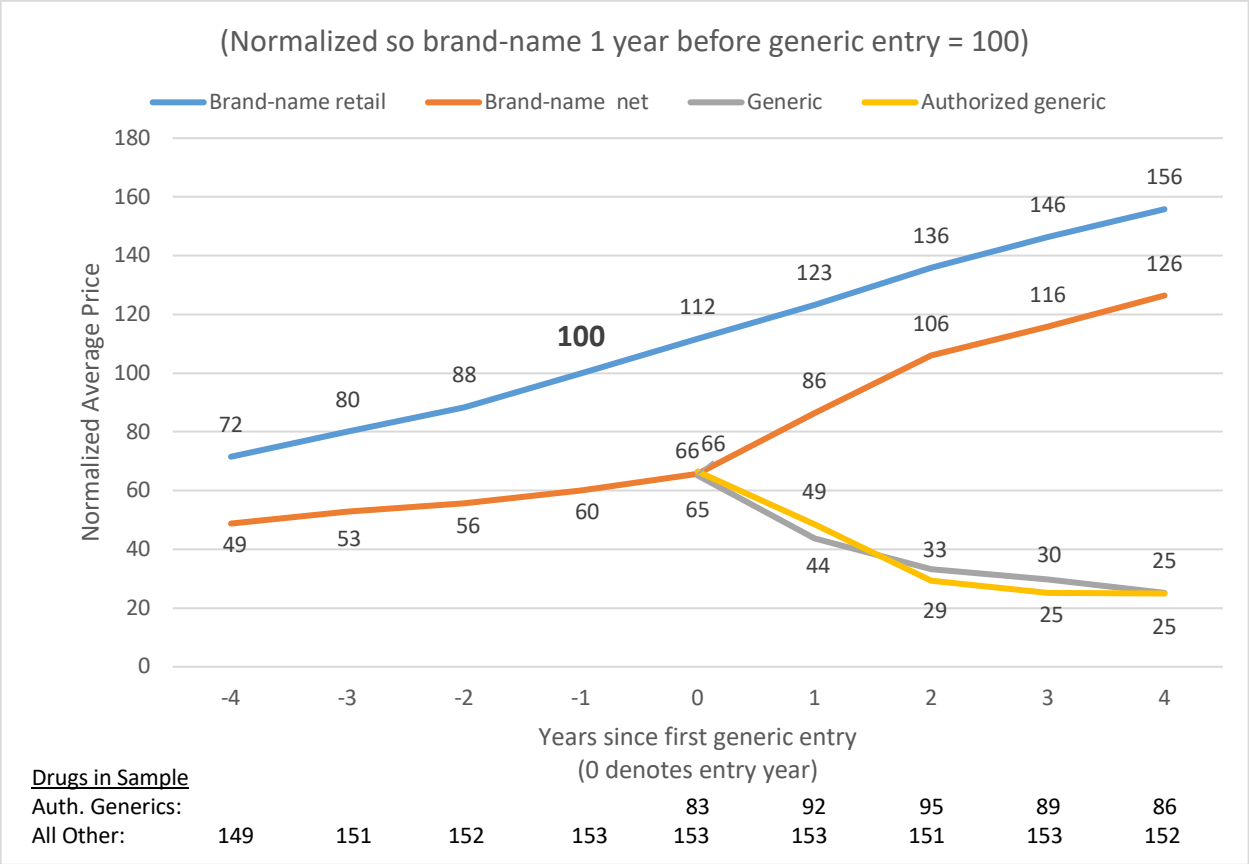
2.4: Results Using Drugs Experiencing First Generic Entry Between 2014 and 2020

We are able to calculate net price and net spending measures for the 2010 to 2024 period, and our sample includes drugs that experienced first generic entry between 2011 and 2023. As a result, there are differences in the set of drugs used to calculate averages at each point along the horizontal axes of the figures, since not every drug in the sample is observed for a full four years before, and four years after, first generic entry. For example, the drugs whose first generic entry was in 2011 are used in the calculations of average prices, prescriptions, and spending for the years denoted -1, 0, 1, 2, 3, and 4 on the horizontal axis, but cannot be used to calculate averages for years -2, -3, and -4. On the other hand, drugs experiencing first generic entry in 2023 are included in the calculations for years -4, -3, -2, -1, 0, and 1, but not 2, 3, or 4.

In this section, we present results estimated using the subset of drugs for which we observe outcomes for a full four years before first generic entry and a full four years afterwards. These are the drugs (153 in total) whose first generic entry occurred between 2014 and 2020.

The results of this subsample analysis broadly align with the results from the full sample. A notable difference is the estimated change in prescription volume following first generic entry: as estimated using the complete sample, total prescription volume increased 49 percent over the four years following generic entry, but as estimated using this subsample, the increase was 22 percent.

Figure S-7: Average Aggregate Prices Among Drugs with First Generic Entry 2014-2020



Drugs in Sample									
Auth. Generics:					83	92	95	89	86
All Other:	149	151	152	153	153	153	151	153	152

Figure S-8: Standardized Prescriptions Among Drugs with First Generic Entry 2014-2020

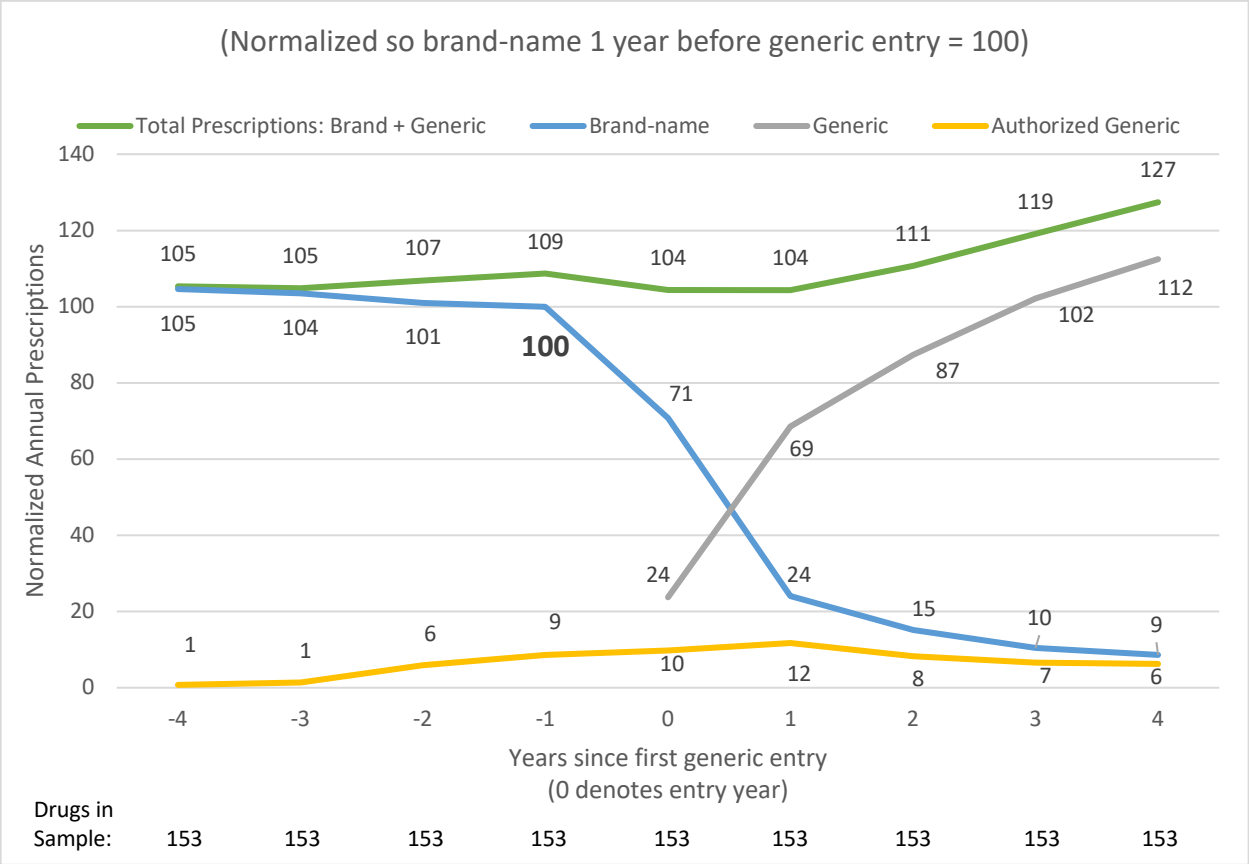


Figure S-9: Part D Net Spending Per Drug Among Drugs with First Generic Entry 2014-2020

